

Ternary or Binary? A Noise-Dependent Benefit Window for Memory-Side Data Governance in Edge Perception

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Abstract—Memory-side data governance promises to relieve interconnect congestion in edge perception by filtering low-utility sensor data before it consumes bandwidth. Beyond binary admit/drop filtering, ternary schemes add a *defer* class whose admission is decided by temporal-redundancy verification in a watchlist buffer. This letter asks a question prior work leaves open: *when does the ternary mechanism actually pay off?* Using a trace-driven simulation on real parking-lot surveillance video (VIRAT), with the deferred class resolved by an explicitly implemented persistence check and with all promoted traffic charged its full transmission cost, we characterize the answer as a *noise-dependent benefit window*. Under reliable observations ($\sigma \leq 0.05$), deferral is pure overhead and binary filtering strictly dominates (AUC difference -0.019 ± 0.001 , $n=10$ seeds). The advantage reverses near $\sigma^* \approx 0.07$, peaks at $\sigma = 0.15$ ($+0.029 \pm 0.004$, $p = 1.3 \times 10^{-7}$), and decays as noise overwhelms temporal verification. Short verification windows dominate throughout ($N=2$); $N=5$ never becomes beneficial. These results yield a concrete deployment rule for memory-side governance hardware: enable ternary deferral only when observation uncertainty is high, and keep the persistence window short.

Index Terms—Edge AI, memory-side governance, data filtering, ternary classification, bandwidth–accuracy trade-off.

I. INTRODUCTION

EDGE perception pipelines stream continuous sensor data toward compute resources across bandwidth-constrained interconnects. Indiscriminate “blind streaming” wastes scarce PCIe and CXL bandwidth on redundant background content, while deferring filtration to host-side software incurs interrupt and cache-pollution overheads [1], [2]. These pressures motivate *memory-side governance*: filtering logic placed at the memory boundary, ahead of both the interconnect and the host OS [3], [4].

The simplest governance policy is a binary threshold: admit a data unit whose estimated utility exceeds τ_U , drop it otherwise. A ternary extension adds a deferred class for ambiguous units, held in a watchlist buffer and promoted only if their salience persists over time. Intuition suggests deferral should help when single observations are unreliable—but it also adds metadata overhead, buffering, and late transmissions whose cost must be paid in full. Whether, and *when*, the ternary mechanism is worth this cost has not been quantified.

This letter makes that question its sole focus and offers three contributions. (1) A trace-driven evaluation in which

the deferred class is resolved by an *actually implemented* temporal-persistence check—not an assumed recovery rate—and in which promoted traffic is charged full transmission cost. (2) A characterization of the resulting *benefit window*: the ternary scheme underperforms binary filtering under reliable observations, becomes superior beyond a noise threshold $\sigma^* \approx 0.07$, peaks at moderate noise, and decays at extreme noise. (3) A design guideline: the persistence window should be short ($N=2$); longer windows forfeit the benefit entirely.

II. TERNARY MEMORY-SIDE GOVERNANCE

A. Classification and Temporal Verification

Let $p_i \in [0, 1]$ denote the observed salience of data unit i (a video tile in our evaluation). With an upper threshold τ_U and a lower threshold $\tau_B = 0.5 \tau_U$, the governance layer performs a ternary split:

$$\text{class}(i) = \begin{cases} \text{Admit}, & p_i \geq \tau_U, \\ \text{Defer}, & \tau_B \leq p_i < \tau_U, \\ \text{Drop}, & p_i < \tau_B. \end{cases} \quad (1)$$

A deferred unit at frame t enters a watchlist and is *promoted* iff its observed salience remains above τ_B for the next N consecutive frames; otherwise it is discarded. This realizes the temporal-redundancy verification of memory-side governance proposals in hardware-friendly form: per-unit state is one counter, and the check is a streaming comparison.

When $\tau_U < 0.25$ the traffic budget is abundant and deferral cannot add information; the policy therefore disables the deferred class in this regime. This adaptive rule is part of the evaluated algorithm and is reflected in all reported costs.

B. Cost Model

Each admitted or promoted unit incurs bandwidth and compute cost $C_b + C_p$ ($C_b=0.05$, $C_p=0.02$ normalized units per tile); each deferred unit incurs watchlist metadata overhead $0.05 C_b$. Crucially, *promoted units pay the full transmission cost*: the ternary scheme receives no subsidized recoveries.

III. EXPERIMENTAL SETUP

Workload trace. We drive the simulation with a real parking-lot surveillance clip from the VIRAT dataset [5] (scene S_0502; 1920×1080 ; 500 frames). Each frame is partitioned into 64×64 tiles (480 per frame); tile salience

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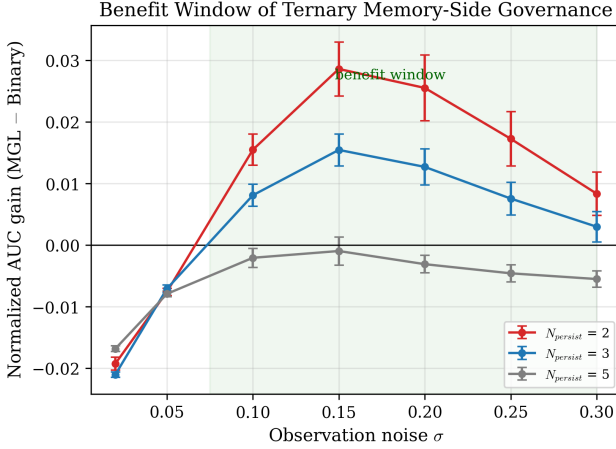


Fig. 1. Noise-dependent benefit window. Paired normalized-AUC gain of the ternary scheme over binary filtering (mean $\pm$ 95% CI, $n=10$ seeds). The gain is negative under reliable observations, crosses zero near $\sigma^* \approx 0.07$, peaks at $\sigma=0.15$, and decays at extreme noise. Longer persistence windows ($N=5$) never become beneficial.

p is the normalized sum-of-absolute-differences against the previous frame.

Task and ground truth. The task is salient-tile recovery: ground truth is the set of tiles whose *clean* saliency exceeds 0.25 (316 tiles over the trace). The governance layer observes $\tilde{p} = \text{clip}(p + \eta, 0, 1)$ with $\eta \sim \mathcal{N}(0, \sigma^2)$, modeling sensor noise, compression artifacts, and front-end estimation error. Recall is the fraction of ground-truth tiles admitted or promoted. We emphasize that this is a trace-driven simulation of the governance mechanism, not a hardware measurement; latency and energy characterization are left to a future FPGA prototype.

Protocol. We sweep $\sigma \in \{0.02, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$, persistence windows $N \in \{2, 3, 5\}$, and ten thresholds $\tau_U \in [0.10, 0.55]$, with $n=10$ random seeds per configuration. Policies are compared by (i) recall–budget Pareto curves and (ii) per-seed normalized area under the recall–budget curve (AUC), with paired differences and 95% confidence intervals across seeds.

IV. RESULTS

A. The Benefit Window

Fig. 1 presents the central result. Under reliable observations the ternary scheme *loses* to binary filtering: at $\sigma=0.02$ the paired AUC difference is -0.019 ± 0.001 (mean $\pm$ 95% CI, $n=10$), because deferral spends metadata and late-transmission cost to confirm what a single observation already reveals. The sign reverses near $\sigma^* \approx 0.07$. The gain peaks at $\sigma=0.15$ with $+0.029 \pm 0.004$ ($p=1.3 \times 10^{-7}$, paired t -test) and remains positive but decaying through $\sigma=0.30$ ($+0.008 \pm 0.004$); once noise corrupts temporal persistence itself, verification promotes spurious tiles and the advantage erodes.

B. Operating Regimes

Fig. 2 contrasts the two regimes at the Pareto level. At $\sigma=0.02$ (panel a) the binary curve dominates across the budget

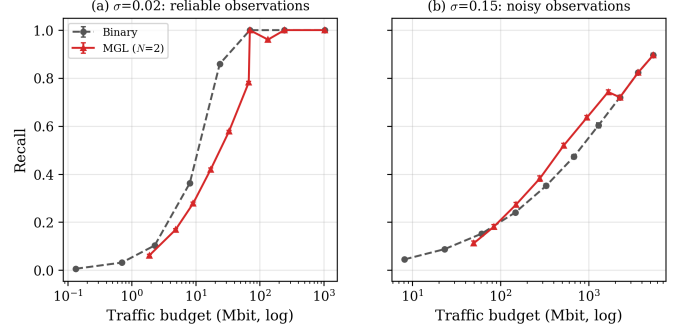


Fig. 2. Recall–budget trade-off in the two regimes ($N=2$; mean $\pm$ SEM, $n=10$). (a) Reliable observations: deferral is pure overhead. (b) Noisy observations: ternary governance dominates.

range. At $\sigma=0.15$ (panel b) the ternary curve lies above binary across mid-to-high budgets: temporal verification recovers genuinely salient tiles that single noisy observations miss, at lower cost than the binary alternative of simply lowering τ_U , which admits noise indiscriminately. The discontinuity in panel (a) near 70 Mbit marks the adaptive defer-disable transition at $\tau_U=0.25$.

C. Persistence-Window Sensitivity

Across the entire sweep, $N=2$ dominates $N=3$, which dominates $N=5$; the $N=5$ gain is never positive (e.g., -0.001 ± 0.002 at the $\sigma=0.15$ sweet spot). Longer verification both delays promotion and raises the bar that noisy persistence must clear, discarding genuinely salient content. For hardware, this is good news: a 2-frame counter suffices, minimizing watchlist state.

D. Limitations

Results derive from a single surveillance trace, a parametric cost model, and additive Gaussian observation noise; the window’s exact boundaries will shift with workload and noise structure, though the qualitative regime structure follows from the mechanism itself. Latency figures of the watchlist path require hardware prototyping and are not claimed here.

V. CONCLUSION

Ternary memory-side governance is not uniformly superior to binary filtering: it occupies a noise-dependent benefit window. Our trace-driven characterization yields a deployment rule for memory-side filtering hardware: under reliable front-end estimates, use a plain binary threshold; enable the deferred class only when observation uncertainty is substantial, and keep the persistence window minimal ($N=2$). These boundaries, obtained with all costs charged honestly, provide the design envelope within which ternary governance hardware is worth building.

REFERENCES

- [1] J. Sim *et al.*, “Computational CXL-memory solution for accelerating memory-intensive applications,” *IEEE Comput. Archit. Lett.*, vol. 22, no. 1, pp. 5–8, 2023.

- [2] Y. Yang *et al.*, “CXLaimPod: CXL memory is all you need in AI era,” *arXiv:2508.15980*, 2025.
- [3] D. Boles, D. Waddington, and D. A. Roberts, “CXL-enabled enhanced memory functions,” *IEEE Micro*, vol. 43, no. 2, pp. 58–65, 2023.
- [4] S. Lee, K. Park, K. Shin, and A. Gavrilovska, “Offloading to CXL-based computational memory,” *arXiv:2512.04449*, 2025.
- [5] S. Oh *et al.*, “A large-scale benchmark dataset for event recognition in surveillance video,” in *Proc. IEEE CVPR*, 2011, pp. 3153–3160.